

DS	Session	Sorting	Question Information	Level 1	Challenge 18
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Question description

kkalaiselvan is going to behave as a vehicle driver.

he needs to drive an automobile on a track divided into "N" no. of sub-tracks.

The letter "K" is also assigned to you.

i.e. the maximum number of kilometres a vehicle may go on each sub-track.

You can add any unit of Petrol to the automobile if it can't cover a sub-track. The total kilometres your automobile may go increases by one unit for every unit of gasoline added.

Input:

The first line of input contains an integer T denoting the no of test cases. Then T test cases follow. Each test case contains two space separated integers N and K.

The second line of each test case contains N space separated integers (A[]) denoting the distance of each N sub-tracks.

Output:

For each test case in a new line you have to print out the minimum unit of Petrol your car require to cover all the sub-tracks. If no extra unit of petrol is required, print -1.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

2
5 6
2 5 4 5 2
5 3
1 6 3 5 2

EXPECTED OUTPUT

-1
3

Test Case 2

INPUT (STDIN)

2
5 6
12 15 14 15 12
5 3
11 61 31 51 21

EXPECTED OUTPUT

9
58

▼ Mandatory Test Cases

Test Case 1

KEYWORD

void sort(int a[],int n)

Test Case 2

KEYWORD

for(i=0;i<n-1;i++)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

5

Test Case 2

TOKEN COUNT

275

Test Case 3

NLOC

40

Sorting8.c +

```
1 #include <stdio.h>
2
3 void sort(int a[], int n) {
4     int i, j, temp;
5     for (i = 0; i < n - 1; i++) {
6         for (j = 0; j < n - i - 1; j++) {
7             if (a[j] > a[j + 1]) {
8                 temp = a[j];
9                 a[j] = a[j + 1];
10                a[j + 1] = temp;
11            }
12        }
13    }
14}
15
16int main() {
17    int t;
18    if (scanf("%d", &t) != 1) return 0;
```

main | CH.SC.U4CSE25225 | Sorting8.c

Ln 36, Col 1 Spaces: 4 UTF-8 CRLF C

```
20    int n, k;
21    scanf("%d %d", &n, &k);
22    int a[1005];
23    for (int i = 0; i < n; i++) {
24        scanf("%d", &a[i]);
25    }
```

```
26     sort(a, n);
27     int max_dist = a[n - 1];
28     if (max_dist <= k) {
29         printf("-1\n");
30     } else {
31         printf("%d\n", max_dist - k);
```

```
32     }  
33 }  
34 return 0;  
35}  
36
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

2

5 6

2 5 4 5 2

5 3

1 6 3 5 2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting8.exe

-1

3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

2

5 6

12 15 14 15 12

5 3

11 61 31 51 21

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting8.exe

9

58

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>